

Skeletal health in adult patients with classic galactosemia.

SUMMARY: This study evaluated bone health in adults with galactosemia. Associations between bone mineral density (BMD) and nutritional and biochemical variables were explored. Calcium level predicted hip and spine BMD, and gonadotropin levels were inversely associated with spinal BMD in women. These results afford insights into management strategies for these patients. **INTRODUCTION:** Bone loss is a complication of galactosemia. Dietary restriction, primary ovarian insufficiency in women, and disease-related alterations of bone metabolism may contribute. This study examined relationships between clinical factors and BMD in patients with galactosemia. **METHODS:** This cross-sectional sample included 33 adults (16 women) with classic galactosemia, mean age 32.0 ± 11.8 years. BMD was measured by dual-energy X-ray absorptiometry, and was correlated with age, height, weight, fractures, nutritional factors, hormonal status, and bone biomarkers. **RESULTS:** There was a significant difference in hip BMD between women and men (0.799 vs. 0.896 g/cm², $p = 0.014$). The percentage of subjects with BMD-Z < -2.0 was also greater for women than men [33 vs. 18 % (spine), 27 vs. 6 % (hip)], and more women reported sustaining fractures. Bivariate analyses yielded correlations between BMI and BMD-Z [at the hip in women ($r = 0.58$, $p < 0.05$) and spine in men ($r = 0.53$, $p < 0.05$)]. In women, weight was also correlated with BMD-Z ($r = 0.57$, $p < 0.05$ at hip), and C-telopeptides ($r = -0.59$ at spine and -0.63 hip, $p < 0.05$) and osteocalcin ($r = -0.71$ at spine and -0.72 hip, $p < 0.05$) were inversely correlated with BMD-Z. In final regression models, higher gonadotropin levels were associated with lower spinal BMD in women ($p = 0.017$); serum calcium was a significant predictor of hip ($p = 0.014$) and spine ($p = 0.013$) BMD in both sexes. **CONCLUSIONS:** Bone density in adults with galactosemia is low, indicating the potential for increased fracture risk, the etiology of which appears to be multifactorial.